The *[general]* section is a standard section that appears at the top of the configuration file for all channel modules, containing general configuration options for how that protocol relates to your system and can be used to define default parameters as well. In Asterisk, we define all actions based on context. In general, we specify that the default context is the context called 'unauthenticated', with no actions defined in this context. So, if users have not authenticated themselves in a context, they will not be able to carry out any action. The *allowguest* parameter specifies that any unauthenticated users are not allowed. Unethical hackers try to guess usernames by trying different usernames. By specifying *alwaysauthreject*, we instruct the system to output 'userrejected' for wrong username+password combinations, and not the 'user not found' message.

In the next section, we define a template we have chosen to name *[test-phone]*. We've created it as a template so that we can use the values within it for all of our devices.

In the *[test-phone]* template, we've defined several options required for authentication and control of calls to and from devices that use that template. The first option we've configured is the type, which we've set to *friend*. This tells the channel driver to attempt to match calls on the name first, and then on the IP address.

The *host* option is used when we need to send a request to the telephone (such as when we want to call someone). Asterisk needs to know where the device is on the network. By defining the value as *dynamic*, we let Asterisk know that the telephone will tell us where it is on the network instead of having its location defined statically.

When a request from a telephone is received and authenticated by Asterisk, the requested extension number is handled by the dialplan in the context defined in the device configuration; in our case, the context is named *office-device*.

The password for the device is defined by the secret parameter. While this is not strictly required, you should note that it is quite common for unethical hackers to run phishing scripts that look for exposed VoIP accounts with insecure passwords and simple device names (such as a device name of 666 with a password of 666). By utilising an uncommon device name such as a MAC address, and a password that is a little harder to guess, we can significantly lower the risk to our system should we need to expose it to the outside world.

Now that we're finished with our template, we can define our device names and, using the *test-phone* template, greatly simplify the amount of details required under each device section. The device name is defined in the square brackets, and the template to be applied is defined in the parentheses following the device name. We can add additional options to the device name by specifying them below the device name.

Each time we change this file, we need to reload the file at the Asterisk prompt. To start Asterisk in verbose mode, use the following command:

```
asterisk -vvvvr
```

The number of 'v's indicate the level of verbosity required. You will obtain the *CLI* prompt. To reload SIP configurations, use the following command:

```
*CLI>sip reload
```

You may verify if the new channels are loaded, by typing the following command:

```
*CLI>sip show peers
```

You would have noticed that Suresh and Ramesh do not have any numbers allocated to them. This is done in the dialplan, by editing */etc/asterisk/extensions.conf*.

```
[office-device]
exten => 100,1,Dial(SIP/ramesh)
exten => 101,1,Dial(SIP/suresh)
```

This basic dialplan will allow you to dial your SIP devices using extensions 100 and 101. If somebody dials the number 100 in the 'office-device' context, the call will be routed to Ramesh. All of these extensions are arbitrary numbers and could be anything you want. You may also choose 4- or 5-digit numbers. You will need to reload your dialplan before changes take effect in Asterisk.

```
*CLI>diaplan reload
```

You should now be able to dial between your two new extensions. Open up the CLI in order to see the call progression. Before that, you need to set up the new phones with the user name and password defined in *sip.conf*. It will show something like what follows:

```
Connected to Asterisk 1.8.23.0-vici currently running on
localhost
Versbosity is atleast 21
[Oct 9 18:11:30] == Using SIP RTP CoS mark 5
[Oct 9 18:11:30] -- Executing [100@office-device:1] Dial("SIP/
ramesh-0000005", "SIP/suresh"
[Oct 9 18:11:30] – Called SIP/suresh
[Oct 9 18:11:30] – SIP/suresh-000006 ringing
[Oct 9 18:11:30] – SIP/suresh-000006 answered SIP/ramesh-0000005
[Oct 9 18:11:30] – Locally bridging SIP/ramesh-00000005 and SIP/
suresh-00000006
```

You have made your first successful call between two IP phones. Congratulations! END

By: Devasia Kurian

The author is the founder and CEO of *astTECS, an Asterisk software-based company providing IP PBX, call centre dialler, etc. He can be contacted at *d.kurian@asttecs.com*. Technical queries and support for items described in this article may be addressed to *http://www.asttecs.com/asterisk-forum/*